Jeffrey Walker

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Summary

Controls and autonomy engineer with graduate research spanning nonlinear aircraft dynamics, robust control, reinforcement learning, state estimation, and simulation-backed evaluation. Strongest evidence comes from a NASA-derived F-18 flight-control program built around controller comparison, rollout analysis, and FlightGear visualization, with additional work in EKF navigation, SLAM, and deployment-oriented embedded systems.

Skills

Controls & Autonomy: LQR, PID, sliding-mode control, nonlinear control, robust control, controller evaluation, reinforcement learning, Monte Carlo robustness analysis

Estimation & Navigation: Kalman filtering, EKF-based sensor fusion, IMU/GPS integration, SLAM workflows, localization error analysis, trajectory alignment

Simulation & Tools: MATLAB, Simulink, FlightGear, MuJoCo, ROS, Gazebo, Python, C/C++, Git, OpenCV

Education

Drexel University M.S. Mechanical Engineering, Robotics, Systems & Controls focus 2023–2025
Thesis: *Advanced Flight Control for Nonlinear Aircraft Models (NASA F-18)* Philadelphia, PA

Kwame Nkrumah University of Science and Technology B.S. Automotive Engineering 2019–2023
Project: *Volkswagen Transporter redesign and electrification* Kumasi, Ghana

Experience

Graduate Research Assistant – Aircraft GNC & Autonomy *Drexel University, Philadelphia, PA* Jun 2024 – Present

- Modeled and analyzed a 12-state nonlinear NASA-derived F-18 aircraft system across multiple operating regimes for advanced flight-control research.
- Designed and evaluated classical and learning-based controllers including LQR, sliding-mode control, robust-control variants, and reinforcement learning agents.
- Benchmarked controller behavior under disturbance, sensor noise, delays, and actuator saturation, with evaluated runs showing up to **61% reduction in peak state excursions** across primary stability-related states.
- Built validation workflows in MATLAB/Simulink and FlightGear to study closed-loop behavior, visual proof, and simulation-to-deployment considerations.

Aerospace Controller Design Researcher *Drexel University – Prof. Bor Ching Chang, Philadelphia, PA* Jul 2024 – Present

- Supported aerospace controls research involving robust controller design with H_2 and H_∞ methods.
- Contributed to MATLAB/Simulink modeling and controller-development workflows for high-precision aerospace applications.
- Helped bridge simulation work toward embedded or hardware-oriented experimentation through Raspberry Pi-based workflows.

Controls and Embedded Systems Intern – Medical Devices *T.I. Inc, Pennsylvania* Sep 2024 – Feb 2025

- Developed embedded control and interface logic for neonatal medical-device subsystems using Arduino, ESP32, C/C++, and Python.
- Integrated sensors, actuators, displays, and motor drivers into electromechanical prototypes with real-time response constraints.

- Contributed to verification and validation activities aligned with **IEC 62304** and **ISO 14971**.

Selected Projects

Nonlinear F-18 Flight Control *Controls / RL / Simulation* 2024 – 2025

- Built a nonlinear aircraft control workflow around a NASA-derived F-18 model using classical control and reinforcement learning methods.
- Organized the project around open-loop versus closed-loop comparison, FlightGear visualization, and technical interpretation instead of abstract AI claims.

Sensor Fusion Navigation *Estimation / EKF / Localization* 2025

- Implemented EKF-based fusion of IMU and GPS signals for ground-vehicle navigation under noisy measurements and intermittent dropout.
- Studied localization behavior across representative trajectories with a focus on robustness and estimation quality.

Stereo SLAM Pipeline for UAV Workflows *SLAM / Perception / Evaluation* 2025

- Built a stereo SLAM workflow with KITTI-style data recording, pluggable backend structure, and trajectory evaluation.
- Structured the project around pose visualization, alignment, and interpretation rather than treating SLAM as a black-box demo.

AeroPendulum Reinforcement Learning Control *Controls / RL / MATLAB-Simulink* 2025 – 2026

- Built aeropendulum control workflows using reinforcement-learning methods and simulation-backed evaluation tooling.
- Compared learning-based behavior against classical control baselines in a dynamic systems setting relevant to controls and autonomy research.

Honors

- **Dean's Fellowship**, Drexel University 2023 – 2025
- **Leroy L. Resser Endowment Scholarship** 2024
- **Master's Thesis:** *F-18 Aircraft Dynamical Control Using Classical Control Methods and Reinforcement Learning: A Comprehensive Study*